Mithan Sakider

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Research Interests

Power System • Renewable Energy • Power System Stability • Power Grid
Green Energy • Signal Processing • Control System • Electric Circuit

Research Summary

Research interests include power system control, load frequency regulation, and artificial intelligence–based optimization techniques. My undergraduate research focuses on optimizing PID controller parameters for two-area Load Frequency Control (LFC) systems using Particle Swarm Optimization (PSO) and Genetic Algorithm (GA). The study evaluates system performance in terms of settling time, overshoot, and steady-state error to enhance frequency stability and overall control performance. MATLAB/Simulink is used for system modeling, simulation, and performance analysis.

Academic Credentials

Bachelor of Science (B.Sc.) in Electrical & Computer Engineering (ECE) January 2020 – December 2026
Rajshahi University of Engineering & Technology (RUET)
CGPA: **3.09/4.00** | Last 60 Credits CGPA: **3.41**

Higher Secondery School Cirtificate (HSC) in Science group 2017 – 2019
Rajshahi Education Board
GPA: **5.00/5.00**

Secondery School Cirtificate (SSC) in Science group 2015 – 2017
Rajshahi Education Board
GPA: **5.00/5.00**

Publications

Research Profiles: IEEE Xplore | ORCID

Conference Proceedings

[1] S. T. Jahan, F. Tabassum and M. Sakider, "AI-Driven Optimization of PID-Based AVR Systems for Enhanced Voltage Regulation in Smart Grids," 2025 International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN), Rangpur, Bangladesh, 2025, pp. 1-6. DOI: 10.1109/QPAIN66474.2025.11172036

On Going Research Project

[1] AI-Driven Optimization Of Two-Area Load Frequency control Systems For Enhanced Frequency Regulation

Research Experience

Research Assistant June 2026 – Present
Supervisor: Dr. Amith Khandakar, Department of Electrical Engineering, Qatar University *Remote*

- Conduct research in machine learning, deep learning, and multimodal learning, contributing to model development, statistical evaluation, reproducible experimentation, and manuscript preparation.

Undergraduate Thesis Research July 2024 – July 2025
Supervisor: Hafsa Binte Kibria, Assistant Professor, Department of ECE, RUET

- Designed ED-Net to address joint Dice-IoU optimization, limited cross-dataset generalization, accuracy-efficiency trade-off, and interpretability gaps in existing polyp segmentation models.
- Proposed a hybrid dual-encoder-decoder (EfficientNetB0 + Double U-Net) with ASPP and SE-block attention, achieving 97.13% accuracy, 92.26% Dice, and 91.08% IoU on Kvasir-SEG with TTA (47.17 FPS without TTA, 12.93 FPS with TTA).
- Demonstrated cross-dataset robustness on three external benchmarks (CVC-ClinicDB: 80.62%, ETIS-LaribPolypDB: 77.15%, PolypGen2021: 80.71% Dice) with a dual-mode clinical deployment strategy for real-time screening and offline diagnostic workflows.

- Conducted Grad-CAM and SE-block attention analysis confirming clinically interpretable decision-making focused on diagnostically relevant polyp regions.
- Published in *Biomedical Signal Processing and Control*, vol. 123, Part A, p. 110566, 2026, Elsevier (Q1, IF: 5.7). DOI: 10.1016/j.bspc.2026.110566

Standardized Test Scores

Language Proficiency (IELTS) 8 November 2025

Overall Band	Listening	Speaking	Reading	Writing
7.0	7.5	7.0	7.5	6.0

Technical Skills

Programming Languages: Python, MATLAB, C, C++, Java.
ML & DL Frameworks: TensorFlow, Keras, PyTorch, Scikit-learn, OpenCV, pandas, NumPy, Matplotlib.
Research Tools: Kaggle, Google Colab, Jupyter Notebooks, Git, GitHub, LaTeX, Mendeley.
Software: Visual Studio Code, Simulink, Arduino, Canva, MS Office Suite.
Web Development: HTML, CSS, JavaScript, PHP, Bootstrap, MySQL, Next.js, MongoDB.

Projects

1. **RUET-ASSR Official Website:** Led a team to build the official website for the Astronomy & Science Society of RUET using Next.js, Prisma, and MongoDB. [Link]
2. **Home Automation System:** Designed an IoT-based home automation system using ESP32 with remote control via Wi-Fi, developed during ECE-3212: Peripherals & Interfacing Sessional. [Link]

Academic Awards and Achievements

1. Technical Scholarship, University Grants Commission (UGC) – three consecutive years: 2022, 2023, and 2024, awarded to top-ranked students based on B.Sc. annual results.
2. General Grade Scholarship in SSC (Grade 10) and JSC (Grade 8) examinations, awarded to nationwide highest-ranked students.

Academic Leadership & Service

1. Chair, IEEE RUET Signal Processing Society (SPS) Student Branch Chapter (2024–2025) [Link]
2. Vice President (Information), Astronomy & Science Society of RUET (ASSR) (2024–2025) [Link]
3. Secretary, Website & CMT Management, 2025 IEEE International Conference on Power, Electrical, Electronics and Industrial Applications (PEEIACON) (2023–2024) [Link]
4. Webmaster, IEEE RUET Student Branch (2023–2024) [Link]
5. Event Management Secretary, Astronomy & Science Society of RUET (ASSR) (2023–2024) [Link]
6. Team Leader, AURORA National Science Festival 2023 (ASSR) [Link]

References

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